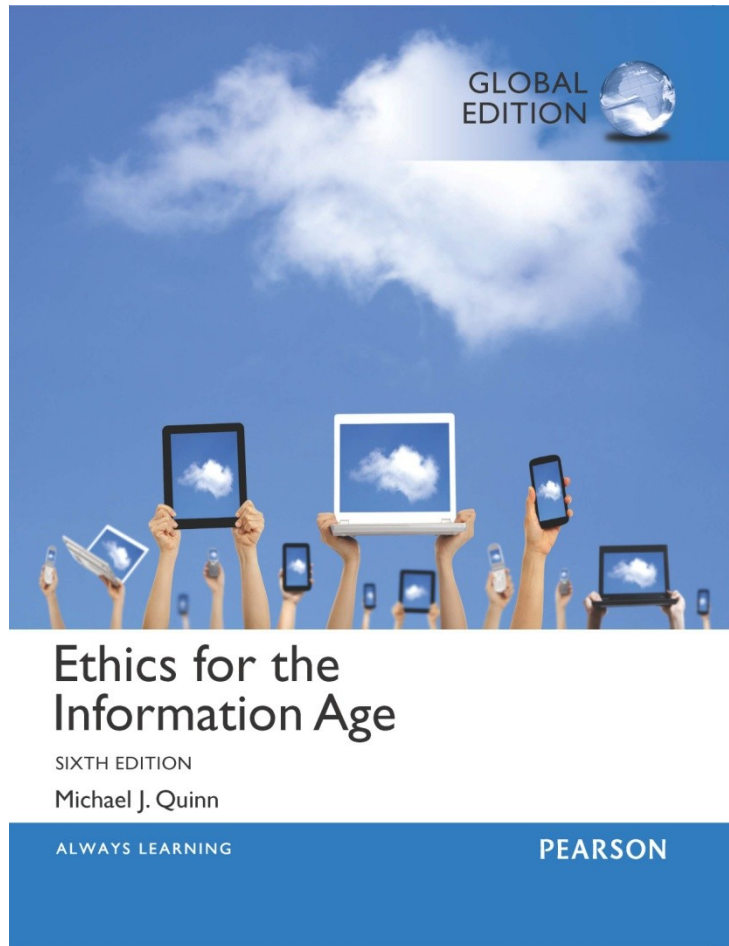




Chapter 2: Networking

Chapter Overview

- Spam
- Internet interactions
- Text messaging
- Censorship
- Freedom of expression
- Children and inappropriate content
- Breaking trust
- Internet addiction

Far More People Have Access to Cell Phones than Electricity or Clean Water



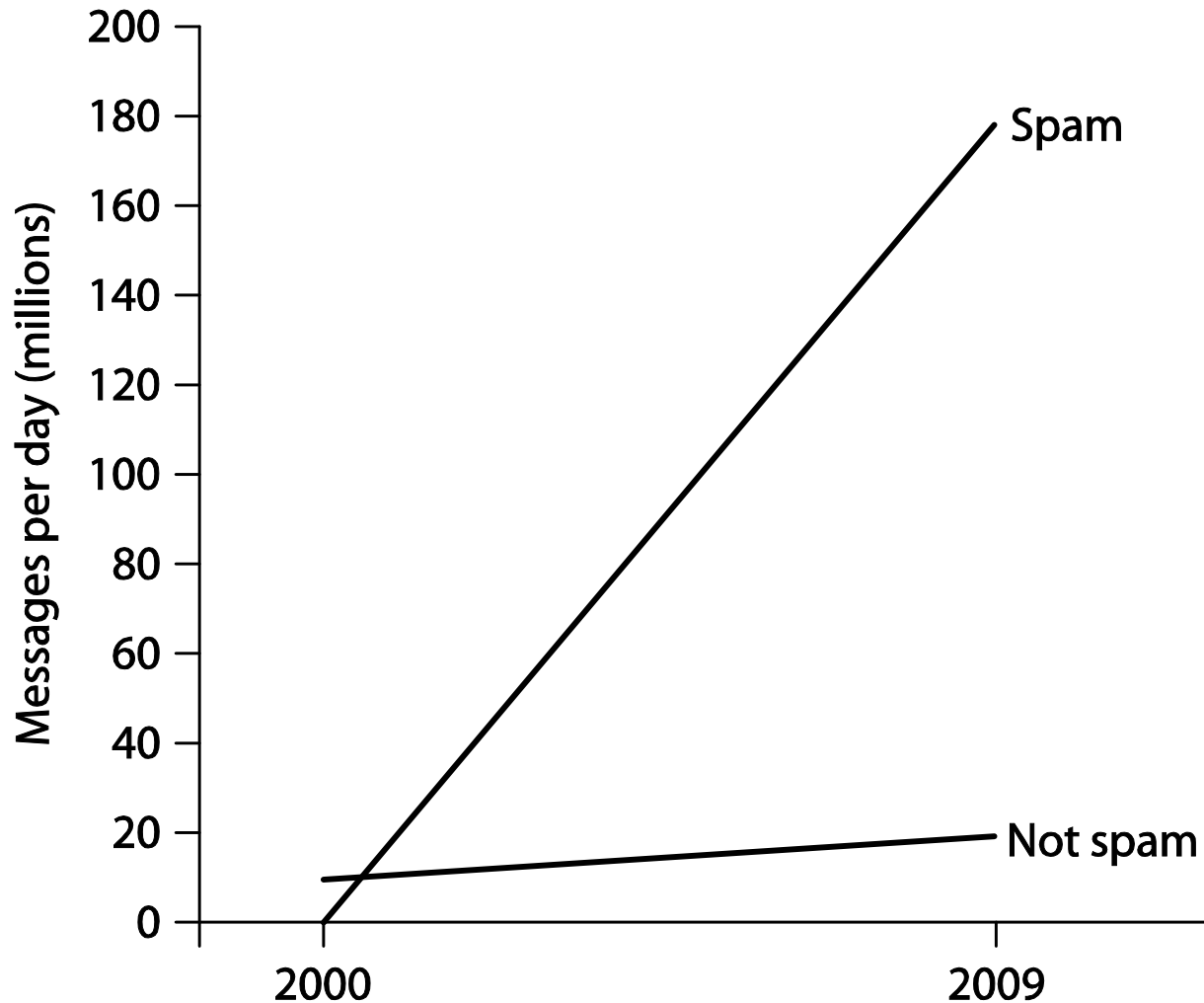
Charles Sturge/Alamy

2.2 Spam

The Spam Epidemic (1/3)

- Spam: Unsolicited, bulk email
- Spam is profitable
 - More than 100 times cheaper than “junk mail”
 - Profitable even if only 1 in 100,000 buys product
- Amount of email that is spam has ballooned
 - 8% in 2001
 - 90% in 2009

The Spam Epidemic (2/3)



The Spam Epidemic (3/3)

- How firms get email addresses
 - Web sites, chat-room conversations, newsgroups
 - Computer viruses harvest addresses from PC address books
 - Dictionary attacks
 - Contests
- Most spam sent out by bot herders who control huge networks of computers
- Spam filters block most spam before it reaches users' inboxes

Need for Socio-Technical Solutions

- New technologies sometimes cause new social situations to emerge
 - Calculators → feminization of bookkeeping
 - Telephones → blurred work/home boundaries
- Spam an example of this phenomenon
 - Email messages practically free
 - Profits increase with number of messages sent
 - Strong motivation to send more messages
- Internet design allows unfair, one-way communications

2.3 Internet Interactions

The World Wide Web

- WWW : networked hypertext system
- Stimulated tremendous growth in popularity of Internet
- Two important attributes
 - It is decentralized
 - Every Web object has a unique address

Rise of the App

- People spending more time on smartphones and tablets
- Using Web browsers awkward on mobile devices
- Mobile apps replace Web browsers as way to access Internet on mobile devices

How We Use the Internet

- Shopping
- Socializing (e.g., Facebook)
- Contributing content (e.g., wikis)
- Blogging
- Crowdsourcing
- Learning
- Exploring our roots
- Entering virtual worlds
- Controlling the Internet of Things
- Paying taxes
- Taking humanitarian action
- Lots more!

2.4 Text Messaging

Transforming Lives in Developing Countries

- M-PESA in Kenya
 - Save money
 - Pay bills
 - Transfer funds
- Communicate information about crop prices
- Protect against counterfeit medicine

Twitter

- Web-based social networking service
- More than 200 million users
- Blogging tool
- Business promotion
- Role in Arab Spring uprisings debated
 - (+) Social networks lead to politicization
 - (-) Ties in social networks too weak for high-risk activism

2.5 Censorship

Forms of Direct Censorship

- Government monopolization
- Prepublication review
- Licensing and registration

Self-censorship

- Most common form of censorship
- Group decides for itself not to publish
- Reasons
 - Avoid subsequent persecution
 - Maintain good relations with government officials (sources of information)
- Ratings systems created to advise potential audience
 - Movies, TVs, CDs, video games
 - **Not** the Web

Challenges Posed by the Internet

- Many-to-many communications
- Dynamic connections
- Huge numbers of Web sites
- Extends beyond national borders, laws
- Hard to distinguish between minors and adults

Governmental Filtering and Surveillance

- Cuba, North Korea: Internet virtually inaccessible
- Saudi Arabia: centralized control center
- People's Republic of China:
 - Blocks Internet access at times of social unrest
 - Has “one of most sophisticated filtering systems in the world”
- Germany: Forbids access to neo-Nazi sites
- United States: Have been repeated efforts to limit access of minors to pornography

Ethical Perspectives on Censorship

- Kant opposed censorship
 - Enlightenment thinker
 - “Have courage to use your own reason”
- Mill opposed censorship
 - No one is infallible
 - Any opinion may contain a kernel of truth
 - Truth revealed in class of ideas
 - Ideas resulting from discourse are more influential

Mill's Principle of Harm

“The only ground on which intervention is justified is to prevent harm to others; the individual's own good is not a sufficient condition.”

2.6 Freedom of Expression

Freedom of Expression: History

- *De Scandalis Magnatum* (England, 1275)
- Court of Star Chamber
- 18th century
 - No prior restraints on publication
 - People could be punished for sedition or libel
- American states adopted bills of rights including freedom of expression
- Freedom of expression in 1st amendment to U.S. Constitution

1st Amendment to U.S. Constitution

Congress shall make no law respecting an establishment of religion, or prohibiting the free exercise thereof; or abridging the freedom of speech, or of the press; or the right of the people peaceably to assemble, and to petition the government for a redress of grievances.

Freedom of Expression Not an Absolute Right

- 1st Amendment covers political and nonpolitical speech
- Right to freedom of expression must be balanced against the public good
- Various restrictions on freedom of expression exist

2.7 Children and Inappropriate Content

Web Filters

- Web filter: Software that prevents display of certain Web pages
 - May be installed on an individual PC
 - ISP may provide service for customers
- Methodologies
 - Maintain “black list” of objectionable sites
 - Examine content for objectionable words/phrases

Child Internet Protection Act

- Libraries receiving federal networking funds must filter pages containing obscenity or child pornography
- U.S. Supreme Court ruled CIPA did not violate 1st Amendment guarantees (6-3 decision in June 2003)

Ethical Evaluations of CIPA

- Kantian evaluation: CIPA is wrong
- Act utilitarian evaluation: depends on how benefits and harms are weighed
- Social contract theory: freedom of conscience should be given precedence

Sexting

- Definition: sexually suggestive text messages or emails with nude or nearly nude photographs
- In a 2009 survey, 9% of U.S. teenagers admitted to sending a sext, 17% admitted to receiving a sext

2.8 Breaking Trust

Identity Theft (1/2)

- Identity theft: When a person uses another person's electronic identity
- Leading form of identity theft is credit card fraud
- Financial institutions contribute to problem by making it easy to open new accounts
- About 8 million cases of identity theft in U.S. in 2010

Identity Theft (2/2)

- Nearly half of cases from lost credit card, checkbook, etc.
- In 20% of cases, credit card number stolen at time of making purchase (skimmers)
- About 1 million cases of identity theft annually in United States from online activities
- **Phishing**: Use of email to attempt to deceive people into revealing personal information
- Identity theft a federal crime, but only 1 in 700 cases results in an arrest

Chat-Room Predators

- Chat room: Supports real-time discussions among many people connected to network
- Instant messaging and chat rooms replacing telephone for many people
- Some pedophiles meeting children through chat rooms
- Police countering with “sting” operations

False Information

- Quality of Web-based information varies widely
- Other media also have information of varying quality
 - *The New York Times v. The National Enquirer*
 - *60 Minutes v. Conspiracy Theory*
- Google attempts to reward quality
 - Ranking uses “voting” algorithm
 - If many links point to a page, Google search engine ranks that page higher

Cyberbullying

- Cyberbullying: Use of the Internet or phone system to inflict psychological harm
- In a 2009 survey, 10% admitted to cyberbullying, and 19% said they had been cyberbullied

2.9 Internet Addiction

Is Internet Addiction Real?

- Some liken compulsive computer use to pathological gambling
- Traditional definition of addiction:
 - Compulsive use of harmful substance or drug
 - Knowledge of its long-term harm
- Some people spend 40-80 hours/week on the Internet, with individual sessions lasting up to 20 hours

Is Internet Addiction Real?

- Kimberly Young created test for Internet addiction
 - Sample question: “Have you repeatedly made unsuccessful efforts to control, cut back, or stop Internet use?”
 - Patients who answer “yes” to at least 5 of 8 questions may be addicted
- Others disagree, noting
 - Computer use is generally considered a positive activity
 - Excessive use does not lead to criminal activity
 - More accurate to call excessive use a compulsion

South Koreans in a PC Bang



Kim-Jae Hwan

1-41

Contributing Factors

- Social factors
 - Peer groups
- Situational factors
 - Stress
 - Lack of social support and intimacy
 - Limited opportunities for productive activity
- Individual factors
 - Tendency to pursue activities to excess
 - Lack of achievement
 - Fear of failure

Ethical Evaluation

- Enlightenment view
 - Individuals can and should govern their lives
 - People are responsible for their choices
- Jeffrey Reiman's view
 - Addict's behavior makes sense if addict has no hope for a better future
 - Society bears responsibility for putting people in hopeless situations